

Internship Project Report: Task 5

IoT Automation with Cloud Control & Mobile App Integration

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Domain: Embedded Systems & IoT

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1. Objective

The objective of Task 5 is to upgrade from one-way sensor monitoring (Task 4) to a **bi-directional IoT control system**. This system enables real-time, remote ON/OFF control of hardware components (LED/Relay) via a Blynk cloud dashboard.

2. Control Signal Flow Explanation

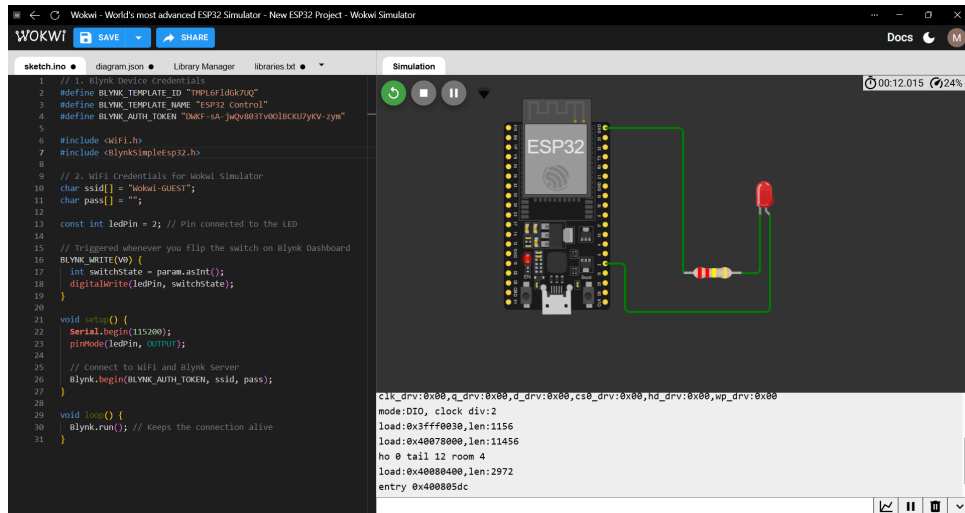
The remote control mechanism operates through a 5-step bi-directional cloud flow:

1. **User Action:** The user toggles the LED switch widget on the **Blynk Web/Mobile Dashboard**.
2. **Cloud Transmission:** The Blynk dashboard sends an API command containing the switch state (1 for ON, 0 for OFF) to the **Blynk Cloud Server**.
3. **ESP32 Reception:** The **ESP32**, connected to Wi-Fi (Wokwi-GUEST), continuously polls the cloud server through the `Blynk.run()` function in the main loop.
4. **Function Execution:** Receiving a state update triggers the `BLYNK_WRITE(V0)` function in the firmware.
5. **Hardware Control:** The ESP32 parses the value (`param.asInt()`) and executes `digitalWrite(ledPin, switchState)`, sending a HIGH or LOW signal to **GPIO Pin 2** to switch the LED state.

3. Circuit Diagram & Hardware Connection

- **Microcontroller:** ESP32 Board
- **Actuator:** Standard Red LED
- **Resistor:** 220 Ω current-limiting resistor
- **Pin Connections:**

- **LED Anode (Positive / Bent Leg):** Connected to **GPIO 2** on ESP32.
- **LED Cathode (Negative / Straight Leg):** Connected through a 220 Ω resistor to **GND** on ESP32.



4. Firmware / Source Code

```

1 // Blynk Template & Authentication Setup
2 #define BLYNK_TEMPLATE_ID "TMPL6FldGk7UQ"
3 #define BLYNK_TEMPLATE_NAME "ESP32 Control"
4 #define BLYNK_AUTH_TOKEN "DWKF-sA-jwQv803Tv00lBCKU7yKV-zym"
5
6 #include <WiFi.h>
7 #include <BlynkSimpleEsp32.h>
8
9 // WiFi Configuration for Wokwi Simulator
10 char ssid[] = "Wokwi-GUEST";
11 char pass[] = "";
12
13 const int ledPin = 2; // Output pin for LED
14
15 // Virtual Pin V0 handler for cloud control signals
16 BLYNK_WRITE(V0) {
17   int switchState = param.asInt(); // Reads 1 (ON) or 0 (OFF) from
18   // Blynk V0
19   digitalWrite(ledPin, switchState); // Sets GPIO 2 output HIGH or LOW
20 }
21
22 void setup() {
23   Serial.begin(115200);
24   pinMode(ledPin, OUTPUT);
25
26   // Establish WiFi & Blynk Cloud Connection
27   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
28 }

```

```

28
29 void loop() {
30   Blynk.run(); // Maintains active communication with Blynk server
31 }

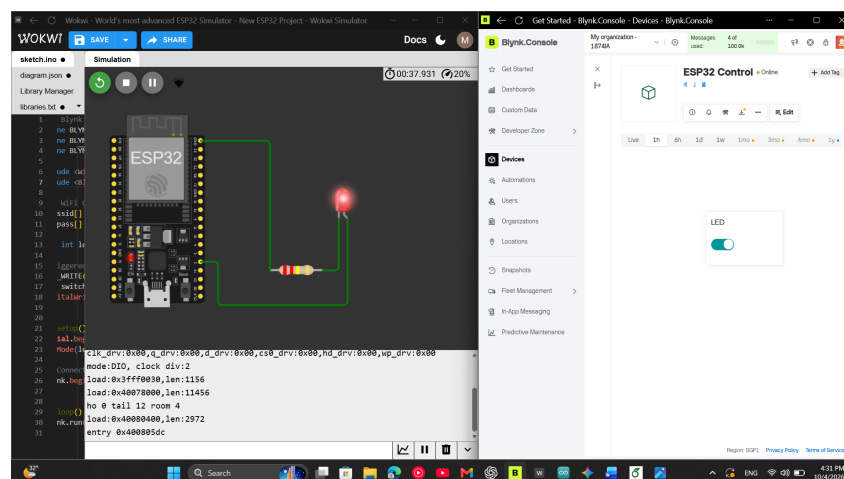
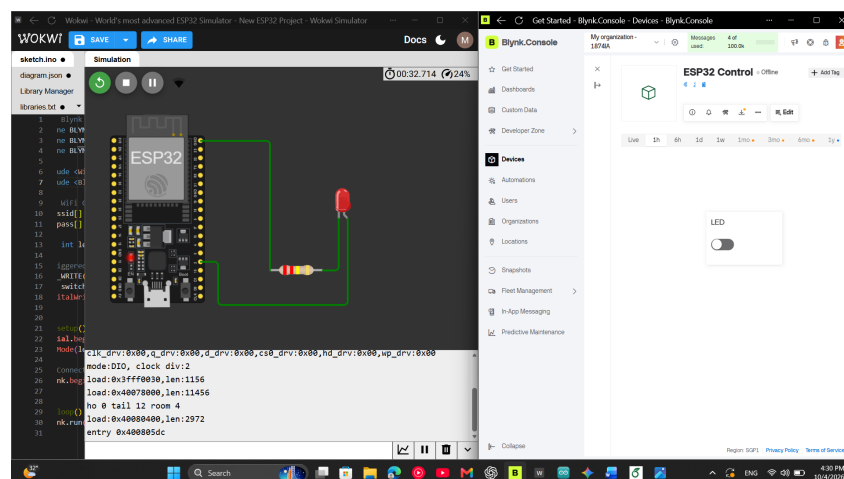
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5. Web / Mobile Dashboard Setup

- **Platform:** Blynk IoT Console (blynk.cloud)
- **Datastream Configured:** Name: LED Control | Pin: Virtual Pin V0 | Data Type: Integer (Min: 0, Max: 1)
- **Widget:** Switch Widget mapped to V0.

6. Verification & System Output

- **Simulation Status:** Verified successfully on Wokwi ESP32 Simulator.
- **Real-time Performance:** Toggling the switch on the Blynk web console turns the simulated LED ON and OFF instantly.



7. Key Learning Outcomes

- Implemented bi-directional IoT communication using Virtual Pins in Blynk.
- Integrated cloud-based remote control with embedded hardware peripherals.
- Configured and verified simulation workflows using Wokwi and Blynk cloud APIs.